

AI3602 Data Mining Homework 2.2

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Question 1

Let $\{i\} \cup C \rightarrow C'$, and similarly for $k_i^{(in)'}$ and other notations.

We have already

$$Q_d(C) = \frac{1}{m} \sum_i \sum_j \left[A_{ij} - \frac{k_i^{(in)} \cdot k_j^{(out)}}{m} \right] \delta(c_i, c_j) = \frac{\Sigma_{(in)}}{m} - \frac{\Sigma_{tot}^{in} \cdot \Sigma_{tot}^{(out)}}{m^2}$$

and $Q_d\{i\} = \frac{1}{m} A_{ii} - \frac{1}{m} \frac{k_i^{(in)} \cdot k_i^{(out)}}{m} = -\frac{k_i^{(in)} \cdot k_i^{(out)}}{m^2}$ is naive.

We'd focus on $Q_d(C')$,

$$Q_d(C') = \frac{1}{m} \sum_i \sum_j \left[A_{ij} - \frac{k_i^{(in)'} \cdot k_j^{(out)'}}{m} \right] \delta(c_i, c_j) = \frac{m \Sigma_{(in)}' - \Sigma_{tot}^{(in)'} \cdot \Sigma_{tot}^{(out)'}}{m^2}.$$

In the above equation, according to $\{i\} \cup C \rightarrow C'$,

$$\begin{aligned} \Sigma_{(in)}' &= \Sigma_{(in)} + k_{i,in} \\ \Sigma_{tot}^{(in)'} &= \Sigma_{tot}^{(in)} + k_i^{(in)} \\ \Sigma_{tot}^{(out)'} &= \Sigma_{tot}^{(out)} + k_i^{(out)}. \end{aligned}$$

Thus,

$$\begin{aligned} \Delta Q_d(i \rightarrow C) &= \frac{m \Sigma_{(in)}' - \Sigma_{tot}^{(in)'} \cdot \Sigma_{tot}^{(out)'}}{m^2} - \frac{\Sigma_{(in)}}{m} - \frac{\Sigma_{tot}^{in} \cdot \Sigma_{tot}^{(out)}}{m^2} + \frac{k_i^{(in)} \cdot k_i^{(out)}}{m^2} \\ &= \frac{\Sigma_{(in)}' - \Sigma_{(in)}}{m} - \frac{\Sigma_{tot}^{(in)'} \cdot \Sigma_{tot}^{(out)'}}{m^2} - \frac{\Sigma_{tot}^{in} \cdot \Sigma_{tot}^{(out)}}{m^2} + \frac{k_i^{(in)} \cdot k_i^{(out)}}{m^2} \\ &= \frac{k_{i,in}}{m} - \frac{(\Sigma_{tot}^{(in)} + k_i^{(in)}) \cdot (\Sigma_{tot}^{(out)} + k_i^{(out)}) - \Sigma_{tot}^{in} \cdot \Sigma_{tot}^{(out)}}{m^2} + \frac{k_i^{(in)} \cdot k_i^{(out)}}{m^2} \\ &= \frac{k_{i,in}}{m} - \frac{k_i^{(in)} \cdot \Sigma_{tot}^{(out)} + k_i^{(out)} \cdot \Sigma_{tot}^{(in)} + k_i^{(in)} \cdot k_i^{(out)}}{m^2} + \frac{k_i^{(in)} \cdot k_i^{(out)}}{m^2} \\ &= \frac{k_{i,in}}{m} - \frac{k_i^{(in)} \cdot \Sigma_{tot}^{(out)} + k_i^{(out)} \cdot \Sigma_{tot}^{(in)}}{m^2}. \end{aligned}$$

Original Questions

Q1. Modularity for Directed Graphs

Define the modularity $Q_d(C)$ for a community C in a *directed graph* G as

$$Q_d(C) = \frac{1}{m} \sum_i \sum_j \left[A_{ij} - \frac{k_i^{(in)} \cdot k_j^{(out)}}{m} \right] \delta(c_i, c_j) = \frac{\Sigma_{in}}{m} - \frac{\Sigma_{tot}^{(in)} \cdot \Sigma_{tot}^{(out)}}{m^2}$$

where

- m is the **number of all links in the graph** G ,
- A_{ij} represents the weight of the edge between node i and j ,
- $k_i^{(in)}, k_i^{(out)}$ are the in-degree and out-degree of node i ,
- c_i is the community to which node i belongs,
- $\delta(c_i, c_j) = 1$ if i and j are in the same community and 0 otherwise,
- Σ_{in} is the sum of edge weights between nodes **inside** the community C ,
- $\Sigma_{tot}^{(in)}, \Sigma_{tot}^{(out)}$ are the sum of **all in- and out-edge weights** for nodes in the community C (including edges which link to other communities).

Prove that the modularity gain $\Delta Q_d(i \rightarrow C)$ of moving node i into community C in the directed graph G is given by

$$\Delta Q_d(i \rightarrow C) = \frac{k_{i,in}}{m} - \frac{k_i^{(out)} \Sigma_{tot}^{(in)} + k_i^{(in)} \Sigma_{tot}^{(out)}}{m^2},$$

where $k_{i,in}$ is the sum of all edge weights between node i and nodes in community C .

Hint. $\Delta Q_d(i \rightarrow C) = Q_d(\{i\} \cup C) - [Q_d(\{i\}) + Q_d(C)]$.

Figure 1: Question 1

Reference

None.